PROJECT AND TEAM INFORMATION

Project Title

StudyStruct: Smart Study Planner Based on Data Structures

Student / Team Information

Team Name: *Team #*	*studystruct*
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Students have to study many subjects and nowadays they usually start preparing only at the last moment Which makes it difficult to decide/prioritize their tasks/subjects, which is more important. Making a study timetable manually can also take a lot of time and can be confusing too. So we thought of making a simple study planner which will help students arrange their tasks based on difficulty and priority.

State of the Art / Current solution (1 pt)

Currently , Students depends on multiple applications such as calender,note keeping apps and planners to study effective with their syllabus and tasks in order . These applications may simplify the tasks arrangements but the students need to prioritise their time . Most applicable lack the ability to management different subjects and topics. This software will help in understanding the implementation to Data structures concepts such as priority queue, graphs, sorting and stack.

Project Goals and Milestones (2 pts)

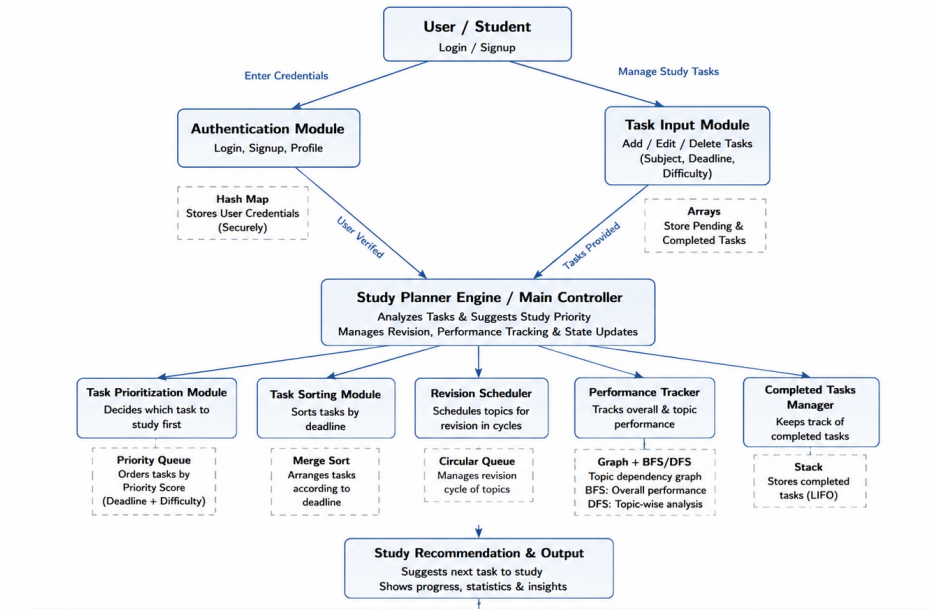
Project goals - 1. To allow students to easily delete, add and update their tasks. 2. To help students decide which task they should prioritize. 3. To arrange the tasks according to their difficulty level and deadline. 4. To use DSA concepts to arrange the tasks. 5. To help students to keep track of their performance, both overall and topic wise and help them to find out which topics are going well and which need more improvement. Milestones - 1. First, we will plan the project and what our project will do and what features we can add to it. We will mainly focus on arranging tasks according to their deadline and difficulty level. 2. Next, we will decide how the tasks will be stored and which data structure we will use to store them 3. After that, we will make the basic features so that students will be able to update, view, add and delete their tasks. 4. Then, tasks will be arranged according to their importance and deadline using Dsa concepts. 5. Next, the system will analyse the available tasks and suggest which task the student should prioritize 6. After implementing features, we will test the project and fix any errors or bugs that we find 7. At last, we will combine all features and prepare the project report and presentation.

Project Approach (3 pts)

1. First we will try to understand the problem students usually face while planning their studies study schedule and decide what features our project should have. 2. The students can add their tasks along with details like subject name , deadline and difficulty level. 3. We will use differnt dsa concepts like priority queue to keep important tasks first, circular queue to manage revision, graphs with bfs and dfs to track both, overall and topic performances, hash map for login details, arrays to store remaining and completed tasks,merge sort to arrange tasks according to their deadline and stack to keep track of completed tasks. 4. After that, the system will analyse the tasks on the basis of its difficulty level and deadline and will arrange them in a suitable order which will help the student to prioritize one tasks at a time 5. Based on the order , the system will suggest the students about which task they should prioritize first. 6. Finally we will test the program with various inputs to check if the system is working properly and if we find any error/bug, we will fix it.

System Architecture (High Level Diagram)(2 pts)

System Architecture (High Level Diagram) – Student Study Planner



The architecture follows a modular, data-structure–driven design where the **Study Planner Engine** acts as the central controller between users, tasks, and data structures. The ***Authentication Module*** uses a ***Hash Map*** to store user credentials securely. The ***Task Input Module*** captures study tasks and stores them in ***Arrays***. The engine prioritizes tasks using a ***Priority Queue***, sorts them with ***Merge Sort***, and schedules revisions using a ***Circular Queue***. Performance is tracked using ***Graphs*** with ***BFS*** and ***DFS*** for overall and topic-wise analysis. **Completed tasks** are managed using a ***Stack***. Finally, the system provides personalized study recommendations and insights to help students plan effectively.

Project Outcome / Deliverables (1 pts)

study planner is a software application that will help students manage their academics, work efficiently and manage their time in the best way possible. With the help of this application students will find it easy to arrange their subjects, assignments all in one application. it will also help students in organising and prioritizing the assignments according to their deadlines. they can also track their progress and revisions. With the help of this project we can understand the role of Data structures and algorithm in real life

Assumptions

We have assumed a student-focused application on a standard Windows/Linux system, with basic user login. Users are assumed to provide valid, complete task details like name of the subject, deadline and difficulty level. Using the deadline and difficulty level provided by the user tasks will be prioritised and the recommendations generated by the system will help the students(user) organize their study workload effectively. Study tasks, completed tasks, revision schedules and performance data will be within the application during its use, without requiring advanced cloud-based storage or synchronisation. We assume team members gave introductory familiarity with C++ and basic DSA concepts. The core functionality of the application will be task management, tasks prioritisation, revision scheduling, performance reviewing.

References

1. Research paper 1 Aicademic Planner: the intelligent study buddy 2. research paper 2 Lumera AI Powered Study Planner and Collaborator 3. I neuron for DSA concepts 4. take u forward for graph concepts